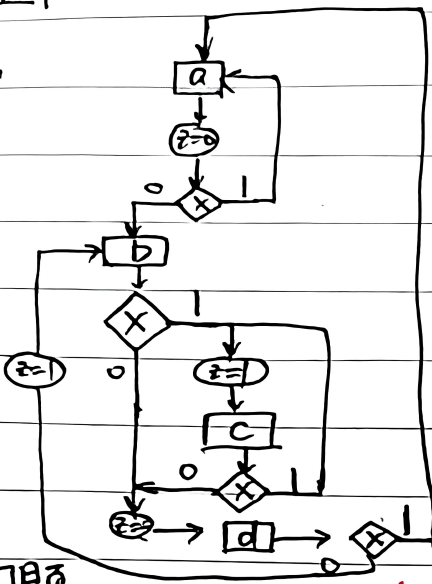


作业10

8.2



补充习题

(1) a b c d e 依次位编码 ~~001 010 011 100~~ . 011 100

用三个 D 触发器 Q_2, Q_1, Q_0

Q_2, Q_1, Q_0	A	Q_2^*, Q_1^*, Q_0^*	Y
0 0 0	0	0 0 0	0
0 0 0	1	0 0 1	0
0 0 1	0	0 1 0	0
0 0 1	1	0 1 1	0
0 1 0	0	0 0 0	0
0 1 0	1	0 1 1	0
0 1 1	0	1 0 0	0
0 1 1	1	0 1 1	1
1 0 0	0	0 0 0	0
1 0 0	1	0 1 1	1

Q_0^*

$Q_0, Q_1 / Q_0, A$	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	X	X	X	X
10	0	1	X	X

$$Q_0^* = A$$

Q_1^*

$Q_2 Q_1 / Q_0 A$	00	01	11	10
00	0	0	1	1
01	0	1	1	0
11	x	x	x	x
10	0	1	x	x

$$Q_1^* = \bar{Q}_1 Q_0 + Q_1 A + Q_2 A$$

Q_2^*

$Q_2 Q_1 / Q_0 A$	00	01	11	10
00	0	0	0	0
01	0	0	0	1
11	x	x	x	x
10	0	0	x	x

$$Q_2^* = Q_1 Q_0 \bar{A}$$

(2)

$$Q_0^* = A$$

$$Q_1^* = \bar{Q}_1 Q_0 + Q_1 A + Q_2 A$$

$$Q_2^* = Q_1 Q_0 \bar{A}$$

$$Y = Q_2 A + Q_1 Q_0 A$$

Y

$Q_2 Q_1 / Q_0 A$	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	x	x	x	x
10	0	1	x	x

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